



Invisible Transformations: Documenting Technical Metadata and Paradata in Web-Based 3D Workflows of Cultural Heritage Objects

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Keywords: 3D workflow provenance; algorithmic processing chains; computational transparency; metadata-driven 3D processing; reproducible digital heritage workflows; research software infrastructure

Abstract

The rapid proliferation of 3D data in the humanities has transformed the documentation, analysis, and communication of cultural heritage objects. Yet the scientific validity and interpretative reliability of such datasets depend not only on the digitised objects themselves but also on the processes used to generate, process, and present them. Consequently, the documentation of metadata and paradata has become essential for transparency and reproducibility in Digital Humanities research.

Current metadata initiatives focus primarily on historical, cultural, and contextual information, while technical conditions of 3D production often remain under-documented (International Organization for Standardization [ISO], 2006; Homburg et al., 2021; Amico & Felicetti, 2021; Storeide et al., 2023). These technical aspects, however, are crucial for assessing the reliability and analytical potential of 3D datasets, particularly when such data are made publicly available and subsequently used by individuals who lack direct knowledge of the digitisation workflow, acquisition parameters, and broader methodological context.

3D models are not neutral reproductions; they reflect specific technological setups and human practices. Factors such as scanning devices, calibration, capture procedures, and operator experience significantly influence the resulting dataset. In our digitisation projects involving historical *Hammerflügel* instruments, scan quality varied considerably depending on both the technical setup and the operator's expertise. As these instruments combine several characteristics that are challenging for 3D acquisition, including reflective surfaces, metallic components, and large homogeneous areas, a detailed understanding of the underlying scanning technologies proved essential for developing reliable and repeatable digitisation workflows (Bosche, 2025).

Post-processing further shapes the analytical validity of 3D models. Steps such as point cloud alignment, mesh generation, and mesh optimisation require methodological decisions and parameter choices that directly affect geometric outcomes. Thus, those processes actively shape the epistemic structure of digital representations (Homburg et al., 2021; Amico & Felicetti, 2021; Storeide et al., 2023, pp. 11-12). Our case studies

demonstrate that even minor variations in mesh-to-mesh alignment can produce significantly different results in deformation analyses intended to investigate structural changes and historical reconstruction processes. For example, altering the subset of mesh data used for the global registration of two scans of the same instrument resulted in substantial differences in the resulting distance maps and, consequently, in the interpretation of changes associated with a reconstruction process.

Challenges are amplified when 3D data are prepared for web visualisation. To ensure interactive performance, models are often simplified or converted to WebGL-compatible formats, distancing the representation from the original measurements. Without systematic documentation of processing steps and parameters, the analytical reliability of web-based models is difficult to assess. These hidden transformations constitute paradata, recording the interpretative and technical decisions embedded in the research workflow. If undocumented, they limit reproducibility and transparency.

From this perspective, we propose understanding 3D digitisation workflows not as sequences of opaque software operations but as chains of algorithmic processes whose assumptions, parameters, and transformations require explicit documentation. We therefore argue for the systematic integration of technical metadata and processing parameters into documentation standards for 3D data, particularly in the context of web-based publication. Such documentation should encompass algorithmic workflows, parameter settings, hardware configurations, and software environments in order to support the assessment of data quality, reproducibility, and analytical validity (Bajena et al., 2022).

At the same time, comprehensive documentation must be balanced against the practical constraints of Digital Humanities projects, including limited time, funding, and technical resources. We address this challenge through the development of DiAuViS SpectraMesh (DiAuViS SpectraMesh, 2026), an extensible software environment that integrates domain-specific requirements into a reusable framework for 3D data processing and publication. In line with FAIR principles, the platform provides open documentation of its architecture, algorithms, and code base, while integrated logging mechanisms record both acquisition contexts and computational processing steps. Although still under active development, the platform has already been deployed in ongoing digitisation projects and demonstrates how technical metadata and paradata can be captured and exposed throughout the 3D workflow.

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